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Advanced Topics in Software Engineering

Set 1: Introduction and Administrivia

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University of Illinois at Chicago

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CS 540: Advanced Topics in Software Engineering

- A seminar-style, active-learning class
  - Introductory lectures given by instructor
  - Student presentations
  - In-class discussions
  - Group project
- Will cover about 20 state-of-the-art research papers in software engineering
- Course topics
  1. Generative AI for software engineering
  2. Software development for hand-held devices (emphasis on Android and security)

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Logistics

- Instructor: Ugo Buy (buy@uic.edu)
  - Office hours: Wednesdays: 3:30-5:30 pm
  - Location: Room SEO 919
- Lecture times: Mon-Wed 11:00 am-12:15 pm
- Lecture location: CDRLC 2407
- Course content: Introductory lectures + Two sets of papers
  1. Papers on Generative AI
  2. Papers on Android and security

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Student responsibilities

- Attend lectures (on campus or online)
  - Attendance is mandatory in an active learning classes
- Give 1 presentation during semester
  - Either on the mobile app development or testing/analysis
- Turn in 2 questions on each paper being presented
  - Identify questions relating to the paper that the paper does not answer
  - Graded with a check ✓ (full credit), ✓+ (extra credit), ✓- (partial credit)
- Participate in discussions during lectures
  - What do you think of papers being presented?
- Group project

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Grading policy

- Presentation – 30 %
- 2-questions + in-class participation – 25%
- Project grade – 25%
- Peer assessment – 20%

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Learning objectives

1. Learn how to read, evaluate and present a research paper
   - What are the research objectives?
   - Did the paper reach its objectives?
   - How innovative?
   - Is the approach broad/narrow?
   - Practical applications?
   - Clear presentation?
2. Learn about applications of generative AI tools to software development
   - Automatic code generation
   - Assisting with requirements writing and traceability
   - Automatic test case generation
   - Applications to software design
3. Tools and techniques for facilitating Android app development

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What is generative AI?

- New branch of AI emerging in Fall 2022
- Predictive AI vs. Generative AI
  - Both use statistical learning methods (i.e., neural nets)
- Predictive AI: Classification (categorical results) and Regression (continuous results)
- Generative AI: Produce new content from Large Language Models (LLM)
- Main inputs to LLMs: Query + Context
- Prompt engineering: Process of defining instructions to guide LLMs to produce desired results

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Why generative AI?

- Tools like ChatGPT, Llama, etc. can be effectively applied to software development
- Andrej Karpathy: “The hottest new programming language is English” (Source: X-posting, January 2023)
  - CS PhD, Stanford University, 2015
  - Co-founder of OpenAI (2015-2017)
  - Director for AI, Tesla (2017-2022)
  - Return to Open AI, GPT-4, (2022-2024)
  - https://karpathy.ai/
- Tremendous potential to enhance productivity of software developers in all phases of the software lifecycle

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Trying out ChatGPT

- Prompt: I was wondering whether you could create a little poem about me
- ChatGPT: **Ugo, a scholar, a mind so bright,**
  Guided by wisdom, reaching new heights.
  At UIC, with passion you strive,
  In the world of knowledge, you truly thrive.
- **Your name carries grace, a story to tell,**
  Of dreams pursued and futures that swell.
  With each step forward, you pave the way,
  Inspiring others, day by day.
- **May your journey be filled with wonder and light,**
  For Ugo, the future is brilliantly bright.

- My prompt: Create a picture of me as a central defender for Inter Milan FC

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Why mobile app development?

- Sales of Android and iOS devices in 2022 were to 60+% of total OS sales, including desktops
  - Going forward, this trend is likely to accelerate
  - See chart next by https://gs.statcounter.com/os-market-share
- The development of mobile apps has unique challenges
  - Platform requirements (e.g., coding by callbacks)
  - Energy constraints
  - Use small screen effectively
  - Protect user's sensitive data (wallet, contacts, calendar, email, browsing history, etc.)
  - Embedded programming with sensors on device (accelerometer, location, thermometer, magnetometer, microphone, etc.)

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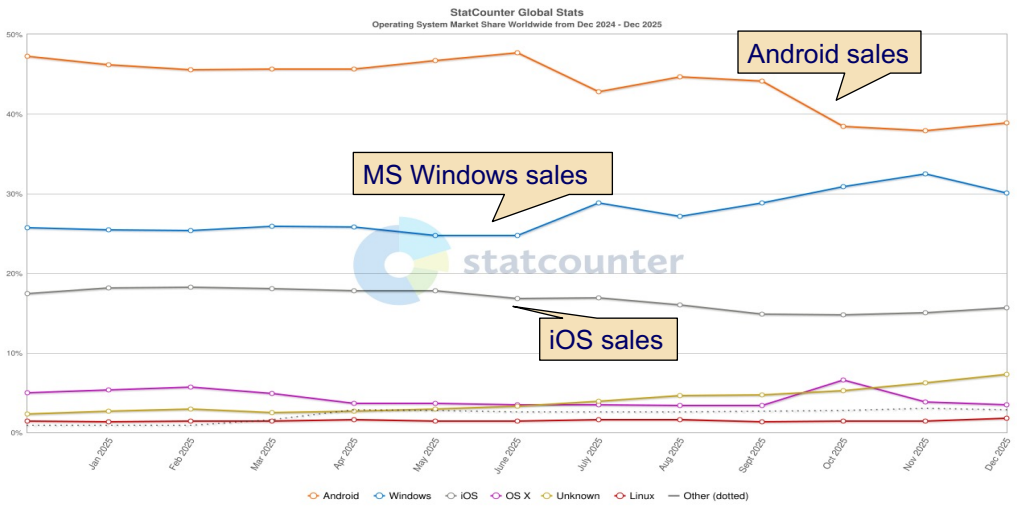
OS sales worldwide, Dec. 2024-Dec. 2025

(Source: https://gs.statcounter.com/os-market-share)



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Why Android?

- Android is a fully open ecosystem
  - OS source code is publicly available
  - Apps can be developed on any desktop platforms (MS Windows, Mac OSX, Linux)
  - Code of apps in Google Play market can often be reverse-engineered to Java-like code
  - Most popular OS
- iOS is quite closed
  - OS source code is (mostly) proprietary
  - Develop apps only on Apple hardware (Macs) with Apple software (XCode)
- Android's openness has made it a target for hackers trying to exploit vulnerabilities (See next slide)
  - Much richer literature on Android than iOS

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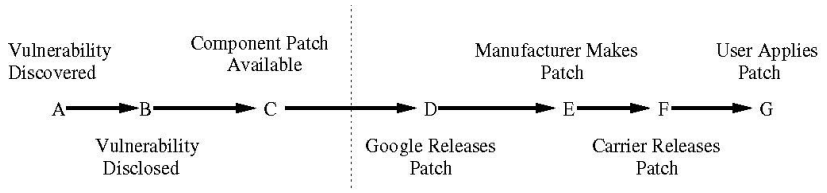
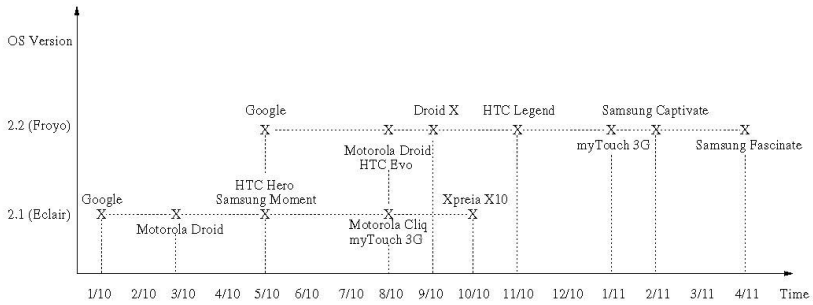
Fragmentation and patch cycles



Source: https://www.usenix.org/conference/woot11/all-your-droid-are-belong-us-survey-current-android-attacks



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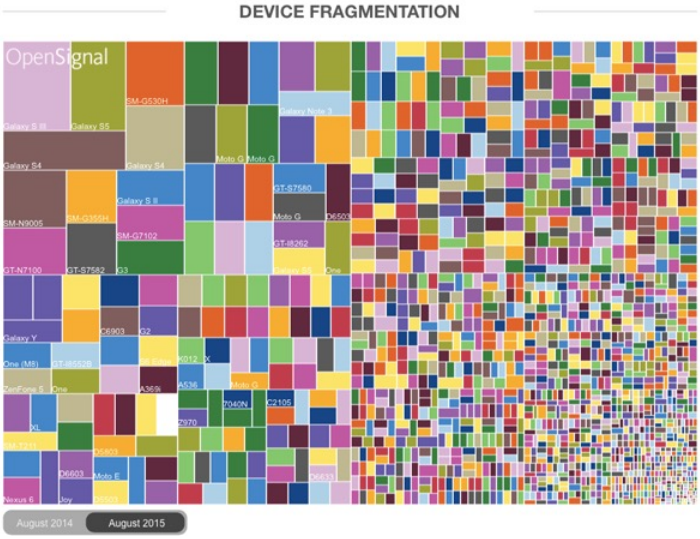
Fragmentation of Android

Source: http://fortune.com/2015/08/05/apple-android-fragmentation/



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Android summary

- High-level overview to familiarize students with relevant portions of Android
- Survey
  - Structure of Android OS
  - App components: (1) Activities, (2) Broadcast Receivers, (3) Services and (4) Content Providers
  - Intents (Explicit vs. Implicit)
  - Foreground vs. background services
  - Security sandbox and permissions

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Selecting the papers

- Top general conferences in the field
  - ACM/IEEE Int. Conference on Software Engineering (ICSE)
  - ACM SIGSOFT Symposium on Foundations of Software Engineering (FSE)
  - IEEE/ACM Int. Symposium on Automated Software Engineering (ASE)
- Top conferences on special topics
  - IEEE Int. Requirements Engineering Conference (RE)
  - ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)
  - Int. Conference on Software Maintenance and Evolution (ICSME)
- Top journals
  - IEEE Transactions on Software Engineering (IEEE/TSE)
  - ACM Transactions on Software Engineering and Methodology (ACM/TOSEM)

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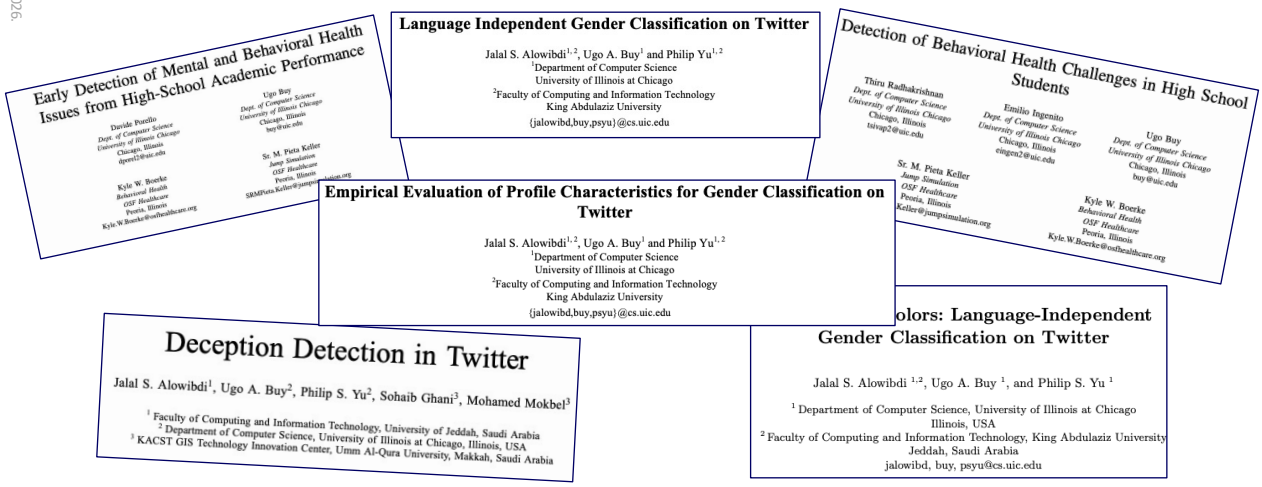
Instructor's background on ML



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Questions

Questions on administrivia?

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